Physiology in the wild: BEATmonitor enables multi-participant PPG recording across naturalistic environments

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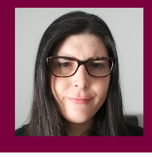


Motivation & overview

Lab-based limitations
- Single participant
- Artificial & short stimuli
- Laboratory environment

Real-world analyses
- Multiple participants
- Natural, long duration stimuli
- Naturalistic environment

BEATmonitor is an open, low-cost framework for multi-participant PPG, heart rate, and accelerometry across live events and everyday settings.

System & workflow

1. Smartwatch Environment

BEATmonitor application hosted on laptop near participants (smartwatches)

Host Computer

Multi-person study

1–120+ Participants

Bangle.js 2 Smartwatch

PPG sensor
Accelerometer

BEATwatch application

[Right] Photoplethysmogram (PPG) provides measure of Cardiac activity (Blood Pulse Volume; BPV)

2. BEATmonitor application

3. User Interface

Web browser interface

Status
- Device name
- Bluetooth
- Connected
- Participant ID
- Ready

Function
- Synchronize time
- Initiate survey question
- Start/stop recording

Transfer
- Search storage
- Transfer progress

Optional client monitors

Network connection

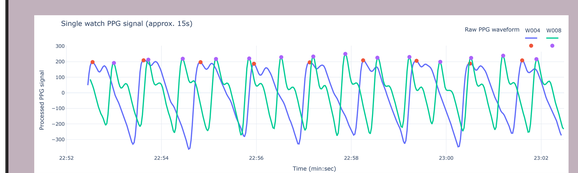
Network Time Protocol (NTP) server

4. Postprocessing script

- Read recorded data files
- Verify recorded data quality
- Trim data
  Resample
- Export processed data for statistical analyses

Identify R-peaks

5. HR/PPG Analysis



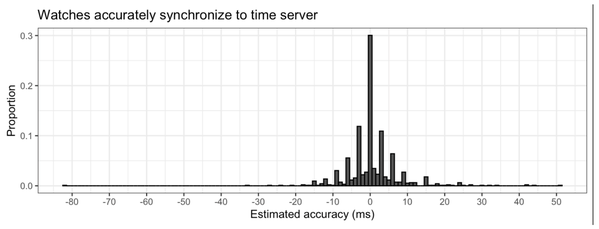
Validation datasets

Data recorded in LIVELab
- Music performance, mime, dance, film screening, debate, art gallery

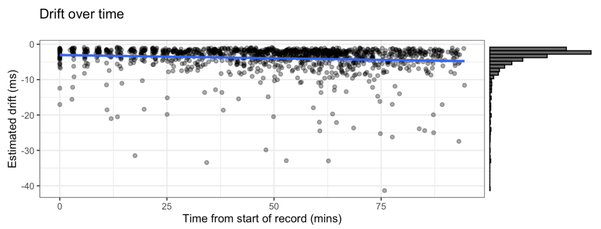
Nine studies / 15 sessions
600+ participants
500+ hours of data

Performance & key findings

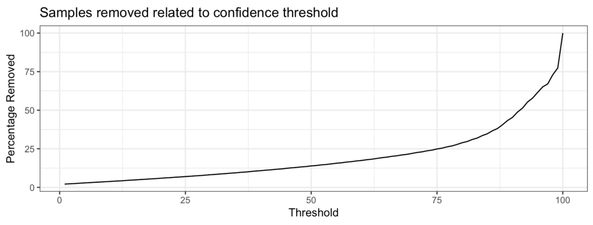
Timestamp error



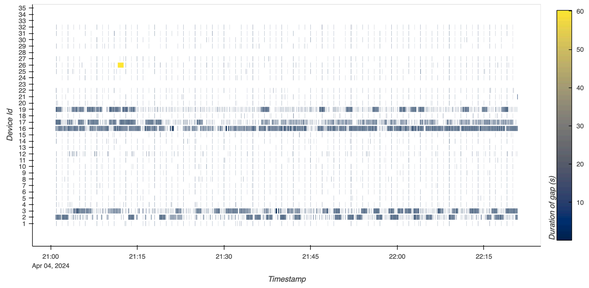
Drift analysis



Confidence threshold



Missing data



PPG / HR

Correlation: Sensor HR vs PPG derived HR

Outcomes

- Psychophysiology in large, naturalistic audiencs.
- Inexpensive, open, scalable to diverse populations and settings.
- Multimodal data (PPG, heart rate, accelerometer, surveys) with synchronized events.